Changyang Li

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Education

George Mason University *Aug 2019 - June 2024*
PhD in Computer Science

University of Virginia *Aug 2017 - May 2019*
MS in Computer Science

Beijing Institute of Technology *Aug 2013 - June 2017*
BS in Computer Science

Skills

Programming	Python, C++, C#, Swift
Deep Learning	PyTorch (DDP/Multi-GPU), Tensorflow
Generative AI	Diffusion Models, Flow Matching, GAN, Graph Generative Model
3D Vision & Reconstruction	3D Gaussian Splatting, Feed-forward 3D Reconstruction, Open3D, OpenCV, OpenGL
Parametric Body Modeling	SMPL-X, MANO, Univeral Hand Model
XR & Game Engines	Unity 3D, Unreal Engine , ARKit, ARCore, RealityKit, Meta Quest, Microsoft HoloLens

Experience

GoerTek Alpha Labs *Feb 2025 - Current*
Research Scientist

- **AIGC**: Utilized diffusion models to synthesize dynamic 3D scenes and human motions.
- **Scene Perception and Visual Localization**: Developed a scene understanding pipeline including object recognition, semantic segmentation, and scene graph modeling techniques for complex indoor environments; Presented a visual localization system by fusing semantic features with geometric constraints.
- **Scene Reconstruction and Segmentation**: Built a feed-forward 3D reconstruction framework with extended features including high-fidelity 3D consistent instance/semantic segmentation, and 3D Gaussian Splatting representations.

Meta *June 2024 - Jan 2025*
Computer Vision Engineer

- **Synthetic Human-Scene Interaction**: Developed an automated pipeline designed to integrates generative human motions along with generative scenes and existing MoCap motion libraries to synthesize highly diverse human-scene interactions.
- **Parametric Human Body/Hand**: Leveraged parametric representations to drive the high-fidelity 3D reconstruction of virtual humans, effectively mapping intricate complex surface geometries to a low-dimensional representation of pose and shape parameter space.

Design Computing and Extended Reality Group *Aug 2019 - June 2024*
Graduate Research Assistant

- **Context-Aware and Interactive XR**: Built adaptive XR systems that dynamically accommodates virtual content based on scene semantics and users' inputs and interactions.
- **XR Spatial Intelligence**: Integrated vision-language models with scene understanding and scene graph modeling to enable virtual characters to navigate and interact with physical environments in XR settings.

Patents

Systems and methods for facilitating navigation in space
Changyang Li, Lap Fai Yu
US Patent 12,254,574, 2025

Selected Publications

Role-Aware Virtual Agents for Navigational Interaction guided by a Multimodal Large Language Model
Minyoung Kim, **Changyang Li**, Cuong Nguyen, Lap-Fai Yu
ACM Transactions on Graphics (Proceeding of SIGGRAPH 2026)

Crafting Dynamic Virtual Activities with Advanced Multimodal Models
Changyang Li, Qingan Yan, Minyoung Kim, Zhan Li, Yi Xu, Lap-Fai Yu
2025 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2025

Synthesizing User-Centered Flying Creature Animation Paths for Outdoor Augmented Reality Experiences
Minyoung Kim, Rawan Alghofaili, **Changyang Li**, Lap-Fai Yu
SIGGRAPH 2024 Conference Papers

Generating Activity Snippets by Learning Human-Scene Interactions
Changyang Li, Lap-Fai Yu
ACM Transactions on Graphics (Proceeding of SIGGRAPH 2023)

Location-Aware Adaptation of Augmented Reality Narratives
Wanwan Li*, **Changyang Li***, Minyoung Kim, Haikun Huang, Lap-Fai Yu
Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems

Optimizing Product Placement for Virtual Stores
Wei Liang, Luhui Wang, Xinzhe Yu, **Changyang Li**, Rawan Alghofaili, Yining Lang, Lap-Fai Yu
2023 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)

Interactive Augmented Reality Storytelling Guided by Scene Semantics
Changyang Li, Wanwan Li, Haikun Huang, Lap-Fai Yu
ACM Transactions on Graphics (Proceeding of SIGGRAPH 2022)

Synthesizing Scene-Aware Virtual Reality Teleport Graphs
Changyang Li, Haikun Huang, Jyh-Ming Lien, Lap-Fai Yu
ACM Transactions on Graphics (Proceeding of SIGGRAPH Asia 2021)

Learning Virtual Grasp with Failed Demonstrations via Bayesian Inverse Reinforcement Learning
Xu Xie*, **Changyang Li***, Chi Zhang, Yixin Zhu, Song-Chun Zhu
2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2019

Earthquake Safety Training through Virtual Drills
Changyang Li, Wei Liang, Chris Quigley, Yibiao Zhao, Lap-Fai Yu
IEEE Transactions on Visualization and Computer Graphics (Special Issue on IEEE VR 2017)

Academic Services and Activities

Program Committee
AAAI, IEEE VR, IEEE ISMAR

Paper Reviewer
- **Conference:** SIGGRAPH, SIGGRAPH Asia, CHI, UIST, Pacific Graphics, IEEE VR, IROS
- **Journal:** ACM Transactions on Graphics, IEEE TVCG, Computer Graphics Forum, IEEE T-RO

Invited Talks
2022 IEEE Metaverse Seminar, 2024 Autodesk ‘Research Connections’ Talk Series